

CO2 ASSESSMENT TOOL 2

ARCHITECTURE DESIGN ANALYSIS REPORT

Project Title

Intelligent E-Governance Citizen Service Request Management System

Course: CSA10 – Software Engineering

Assessment: CO2 Assessment Tool 2 – Architecture Design Analysis Report

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Student Details & Certification

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Code of Conduct:

I, *[S.Suvikumar / 192424262]*, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

1. System Requirement Analysis

1.1 System Objectives

The primary objective of the Intelligent E-Governance Citizen Service Request Management System is to provide a centralized digital platform for submitting, processing, tracking, escalating, and resolving citizen service requests. It replaces fragmented manual processes with an integrated workflow that improves transparency, response time, accountability, and citizen satisfaction.

The platform accepts requests for civic and government services, validates and classifies information, routes requests to the appropriate department, monitors service-level deadlines, sends notifications, records actions, and provides dashboards and reports for officials and administrators.

1.2 Functional Requirements (FR)

- FR-1 Citizen Request Submission: Allow citizens to submit requests through web or mobile interfaces with service category, description, location, and supporting documents.
- FR-2 Authentication & Identity Management: Securely authenticate citizens, officers, supervisors, administrators, and auditors with role-based access.
- FR-3 Intelligent Request Classification: Classify incoming requests using service categories, keywords, configurable rules, and optional AI-assisted classification.
- FR-4 Automated Department Routing: Route requests to the appropriate department and work queue based on category, location, priority, and routing rules.
- FR-5 SLA & Priority Management: Assign configurable priorities and service deadlines and monitor requests against SLA targets.
- FR-6 Request Processing: Allow officers to review requests, request additional information, update status, add remarks, attach evidence, and close completed requests.
- FR-7 Notification Management: Send submission, assignment, status, escalation, and resolution notifications through SMS, email, and push channels.

- **FR-8 Escalation Management:** Escalate overdue or critical requests to supervisors according to configured rules.
- **FR-9 Feedback Management:** Allow citizens to provide ratings and comments after service completion.
- **FR-10 Analytics & Reporting:** Provide dashboards and reports for request volume, response time, SLA compliance, workload, resolution rate, and feedback.
- **FR-11 Audit Logging:** Record significant user, workflow, assignment, status, and configuration changes.
- **FR-12 External Integration:** Provide secure APIs for approved identity, notification, mapping, payment, and departmental systems.

1.3 Non-Functional Requirements (NFR)

- **NFR-1 Latency:** Normal citizen and officer operations should target a response time below 3 seconds under expected load.
- **NFR-2 Scalability:** Support expansion from one department or municipality to multiple departments and regional service units.
- **NFR-3 Availability:** Target 99.9% availability for core request submission, tracking, processing, and notification services.
- **NFR-4 Security:** Enforce encrypted communication, secure authentication, role-based access control, audit logging, and protection of sensitive citizen data.
- **NFR-5 Reliability & Recovery:** Use durable queues, database backups, monitoring, and recovery procedures to reduce data loss and interruption.
- **NFR-6 Accessibility & Usability:** Provide responsive interfaces following WCAG 2.1 Level AA principles.
- **NFR-7 Maintainability:** Use modular services, documented APIs, testing, logging, and configuration-driven workflows.
- **NFR-8 Interoperability:** Support secure API-based integration with approved government and external services.

2. Selection of Software Architecture

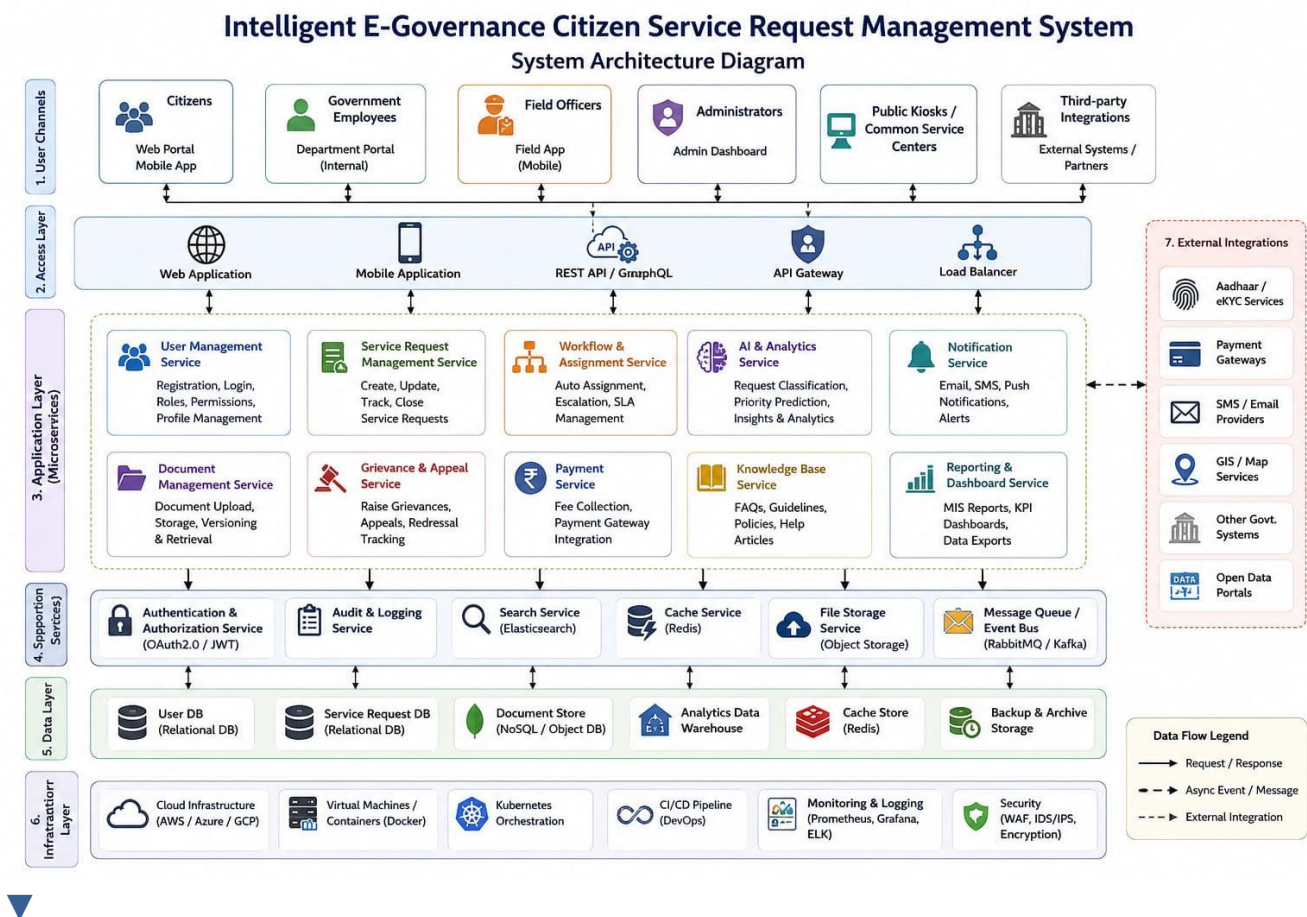
2.1 Selected Architectural Style

Event-Driven Microservices Architecture with API Gateway and Workflow Services

2.2 Architectural Justification

- **High Request Volume:** Event-driven processing handles bursts of citizen requests and background work asynchronously.
- **Independent Scalability:** Request processing, notifications, analytics, and document services can scale independently.
- **Fault Isolation:** Failure of analytics or notification components should not stop core request submission and tracking.
- **Workflow Automation:** A dedicated workflow service manages assignment, SLA timers, escalation, approval, and resolution stages.
- **Security & Governance:** API Gateway and identity services provide consistent authentication, authorization, rate limiting, and auditing.
- **Integration Flexibility:** Service-based APIs support integration with existing departmental and government platforms.

3. Software Architecture Diagram



4. Component Responsibilities and System Workflow

4.1 Component Responsibilities

Component Layer	Purpose & Key Responsibilities
Citizen Web/Mobile UI	Submit requests, upload documents, track status, receive notifications, provide feedback.
Officer/Admin Portal	Process assigned requests, configure services, monitor queues, and view reports.
API Gateway	Routes requests, authenticates users, applies authorization and rate limits.
Identity & Access Service	Manages authentication, roles, permissions, sessions, and optional MFA.

Request Management Service	Maintains request records, statuses, assignments, comments, and resolutions.
Classification & Routing Service	Determines category, priority, department, and queue using rules or AI assistance.
Workflow & SLA Service	Manages workflow stages, deadlines, reminders, escalations, and approvals.
Notification Service	Delivers SMS, email, and push notifications with retry and delivery tracking.
Analytics & Reporting Service	Generates dashboards, KPIs, trends, and downloadable reports.
Database & Document Storage	Stores structured request data, metadata, audit information, and supporting documents.
Audit Service	Records important actions and system events for accountability.

4.2 System Workflow

- **Request Submission:** Citizen selects a service, enters details, uploads documents, and submits.
- **Validation & Registration:** The system validates information, stores the request, and generates a unique request ID.
- **Classification & Routing:** The classification service determines category, priority, and department.
- **Assignment & SLA Start:** The workflow service assigns the request and starts the configured SLA timer.
- **Processing:** The officer verifies information, communicates with the citizen if needed, and updates status.
- **Monitoring & Escalation:** The SLA service sends reminders and escalates overdue requests.
- **Resolution:** The officer records the resolution and closes the request.
- **Notification & Feedback:** The citizen receives completion notification and can provide feedback.

- Analytics & Audit: Events update dashboards, performance indicators, and audit records.

5. Evaluation of Quality Attributes

- Scalability: Stateless services can be horizontally scaled behind the API Gateway, while asynchronous events reduce coupling.
- Security: HTTPS/TLS, secure identity management, RBAC, audit logging, and controlled API access protect data and services.
- Performance: Fast request operations are separated from background notification, reporting, and analytics tasks.
- Reliability: Durable queues, backups, retries, health checks, and monitoring reduce service failure impact.
- Maintainability: Independent services allow components such as notification, classification, and reporting to be updated separately.
- Availability: Redundant service instances and database recovery mechanisms support continuity during isolated failures.
- Usability: Responsive forms, simple service categories, clear status labels, and accessible interfaces reduce citizen effort.

6. Architectural Justification and Technical Trade-Offs

6.1 Rationale

An Event-Driven Microservices Architecture is selected because the platform contains independent workflows for request registration, classification, departmental routing, SLA monitoring, notifications, analytics, and auditing. An event mechanism allows these functions to communicate without making every operation synchronously dependent on all other services.

Compared with a traditional monolithic design, the proposed architecture provides separation of responsibilities, independent scalability, fault isolation, and easier integration with existing departmental systems through secure APIs.

6.2 Trade-Offs Considered

Trade-Off	Decision & Reason
Complexity vs. Decoupling	Microservices increase operational complexity but provide independent scaling and clearer service boundaries.

Consistency vs. Performance	Strong consistency is preferred for identity, permissions, request ownership, and workflow state; eventual consistency is acceptable for analytics and dashboards.
AI Automation vs. Human Control	AI-assisted classification can reduce manual work, while uncertain or sensitive cases remain subject to authorized human review.
Cloud Scalability vs. Cost	Cloud deployment supports elastic scaling but requires resource and cost monitoring.
Centralization vs. Department Integration	A common request platform improves transparency while APIs allow departments to retain specialized systems.

7. Conclusion

The Intelligent E-Governance Citizen Service Request Management System uses an event-driven microservices architecture to support a secure, scalable, reliable, and maintainable citizen-service platform. The architecture separates request processing from classification, workflow, notification, analytics, and audit functions while allowing them to communicate through secure APIs and asynchronous events.

The design supports centralized request management, automated routing, SLA monitoring, notifications, transparent tracking, analytics, and auditability. The principal trade-offs are addressed through clear service boundaries, strong consistency for critical data, human oversight, scalable deployment, and API-based interoperability.